SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT ENGINEERING

Academic Year: 2025-26

Program: B Tech / MBA Tech / B Tech Intg (COMP)

Subject: Microprocessor and Microcontroller | Year: II/IV | Semester : IV / VIII

Date : 09 May 2026 | Time: 02.00 pm - 05.00 pm (3 hrs)

Marks: 100 | No. of Pages: 02

Final Exam 2025-26 / Re-Exam (2022-23, 2023-24, 2024-25)

Instructions:

1. This question paper contains 2 pages.
2. Section A is compulsory.
3. Attempt any 4 questions from Section B.
4. In all 5 questions are to be attempted.
5. All questions carry equal marks.
6. Answer to each new question to be started on a fresh page.
7. The figure on the right-hand side indicates full marks.
8. Assume suitable data if necessary.

SECTION A

Q1		Answer briefly:	[20]
CO- 1; BL-2	a.	Write all combinations of segment registers and offset registers that are used to represent a logical address for each segment in the 8086. What will be the 20-bit physical address of the last memory location of the code segment with CS=1234H? What is the reset address of the 8086 microprocessor?	[05]
CO- 4; BL-4	b.	Explain the function of BUSY and TEST signals in synchronization between the 8086 and the 8087.	[05]
CO- 3; BL-3	c.	Name the I/O ports of the 8051 and describe the characteristics of each.	[05]
CO-4 ; BL-3	d.	What is Arduino? Explain its features. What are the advantages of Arduino in embedded systems?	[05]

SECTION B

Q2 CO-1; BL-2	a	Draw and explain the programming model of the 8086 Microprocessor. Draw the flag register of 8086 and explain all the control flags in detail.	[10]
CO-2; BL-4	b	Write a commented 8086 ALP to perform the following addition 86 9A 42 3C +3B 84 6A 15	[10]

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Q3 CO-2; BL-2	a	What are the different priority modes available in the 8259 Programmable Interrupt Controller (PIC)? Explain each priority mode in detail.	[10]
CO-2; BL-3	b	Explain the architecture of the 8087 Math Coprocessor with a neat block diagram and describe the function of each unit.	[10]
Q4 CO-1; BL-4	a	How is parallel processing achieved in the 8086 Microprocessor? Explain the functions of BIU & EU in detail. Also, explain how pipelining improves performance? Give an example of two instructions that will degrade the performance of pipelining.	[10]
CO-2; SO-; BL-3	b	Write an 8086 ALP to perform the following operations (i) Load the contents of memory locations addressed by SI into AX using a string instruction. (ii) Add 6 & 8.Enter data in unpacked BCD (iii) Subtract 6 from 8. Represent numbers in ASCII	[10]
Q5 CO-; 3,SO-; BL-3	a	With the help of a diagram, explain the Timer/Counter control logic.	[10]
CO-; 3, BL-3	b	Describe the Interrupt Enable register of 8051. Write an 8051 ALP to load the accumulator with the value 55H, and complement the ACC 700 times.	[10]
Q6 CO-2; BL-2	a	Describe types of Hardware & Software interrupts in the 8086 Microprocessor.	[10]
CO-4; BL-4	b	List the applications of advanced microcontrollers. Explain any one application of an advanced microcontroller used in industry.	[10]
Q7 CO-3; BL-2	a	Describe the following special function registers in the 8051. (i) SBUF (ii) PCON (iii) TCON (iv) PSW	[10]
CO-3; BL-4	b	Write a neatly commented 8051 ALP to transmit character 'B' continuously on the serial port at 4800 baud in 8-bit UART mode. Fosc = 11.0592MHz.	[10]

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